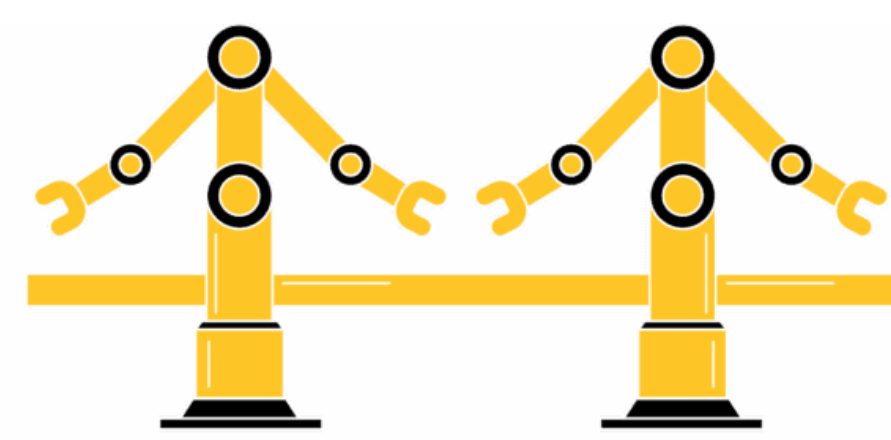




Craters as Constellations: Adapting Star Identification to Lunar Crater Identification

Jeongbin Sohn^{1*}, Hyunsung Kim^{2*}, Pyojin Kim¹, Seokju Lee²

¹Gwangju Institute of Science and Technology, ²Korea Institute of Energy Technology
* Equal contribution



Motivation & Problem

• Why Do We Need to Identify Craters?

- Crater Identification (CID) is a part of Terrain Relative Navigation (TRN), which is crucial for lunar landers due to the absence of GNSS

• Inspiration from Star Identification

- Similarity: existence of pre-made catalog
- Difference:

	Crater	Star
Database Size	1 million	10k
Distance to Objects	Limited	∞
Geometric/Photometric Variety	High	Low

→ CID is harder than SID

• Limitations of the Previous Attempts

- A work utilized scale-invariant features based on crater triangles, but triangle angles and relative edge geometry vary with off-nadir viewing.
- Another work applied the Pyramid algorithm to CID by using projective invariants. Yet, it was highly sensitive to the crater detection noise.

• Key Idea

- Add scale-invariant features to original SID methods!
- SIGNN → CIGNN (Crater Identification via Graph Neural Networks)
- Tetra → CITRA (Crater Identification via teTRA)

Proposed Method

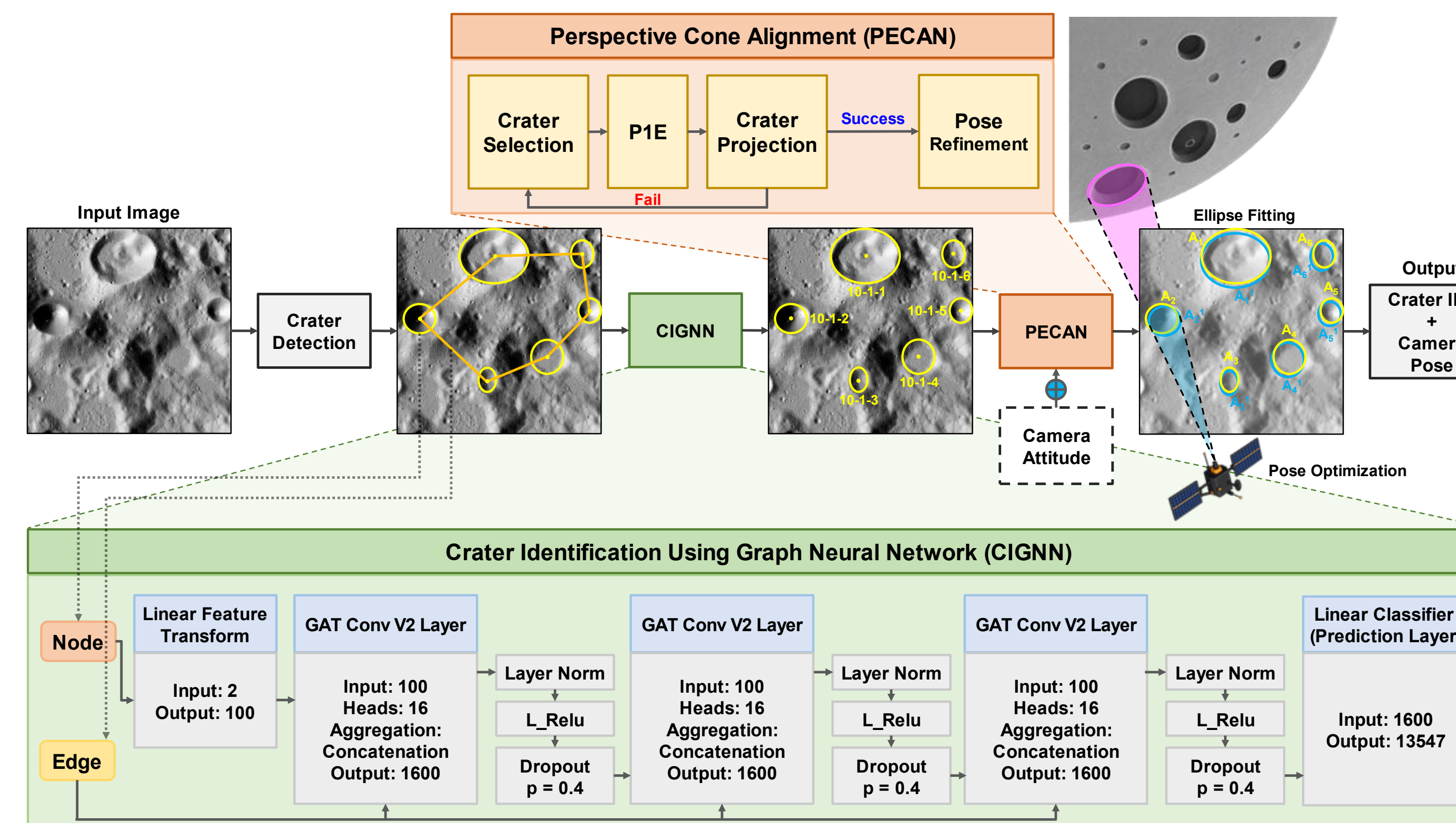


Fig. 1. **CIGNN Workflow**. It inputs a lunar surface image and outputs crater IDs and the camera's pose. We bypass the crater detection step. CIGNN uses a three-layer Graph Attention Network, and it predicts crater IDs with the corresponding confidence scores. Finally, PECAN re-infers crater IDs with one correctly identified crater.

- **Crater-Aware Graph Features**. Set a node feature as a relative area and eccentricity of a crater, and an edge feature as a relative distance.
- **Bounded Runtime**. PECAN after the CIGNN inference takes $O(1)$ rather than original $O(N)$.

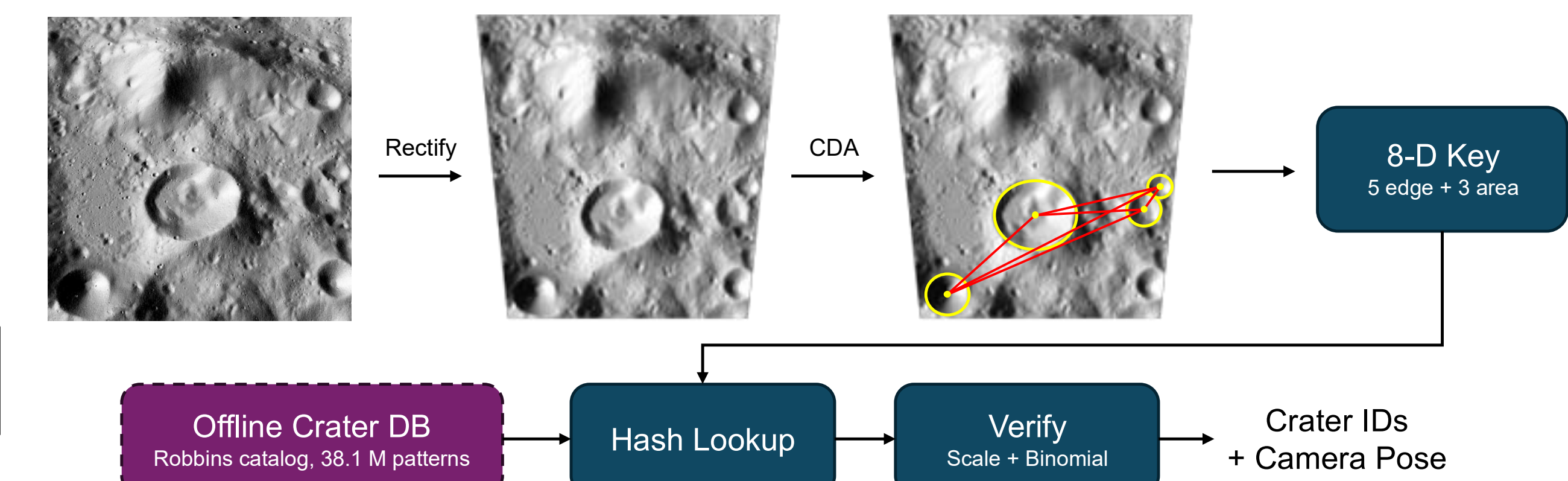


Fig. 2. **CITRA Workflow**. Given a known camera attitude, the input image is first rectified to restore the similarity relation. Each four-crater pattern is encoded as an 8-D key (five edge ratios and three rim-area ratios), looked up in an offline hash table built from the Robbins catalog, and verified by reprojecting the catalog to output crater IDs and the camera's pose.

- **Off-Nadir Rectification**. attitude from IMU + star tracker gives a single homography H ; rims are warped as $C' = H^T C H^{-1}$ (not via their centers)
- **Size-Aware 8-D Key**. three rim-area ratios added and binned coarsely, acting as a retrieval filter against noisy size estimates
- **Bounded Runtime**. only the 8 largest craters form patterns → at most ${}_8C_4 = 70$ lookups per frame

Evaluation & Conclusion

• Evaluation on QuickMap Dataset

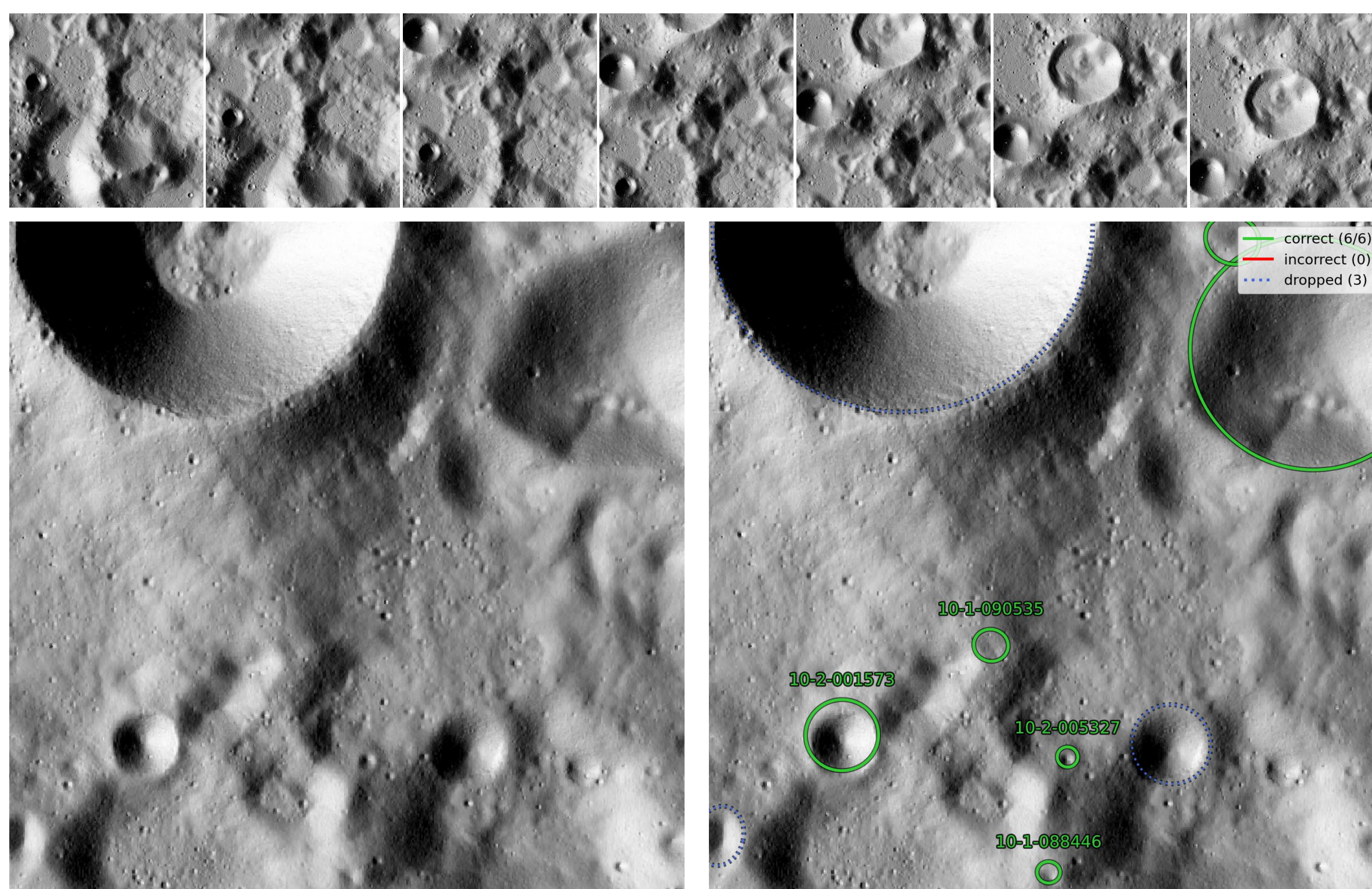


Fig. 3. **Evaluation results of CIGNN and PECAN on a QuickMap image**. The upper sequence is generated using Lunar/LROC QuickMap with off-nadir angle of 0° . The left image is the 50th image of the sequence, and the right image is the inference result of CIGNN and PECAN. The green and red circles indicate correctly identified craters and incorrectly identified craters, respectively. Circles do not fit to the imaged craters because their elevations are not considered in projection.

Method	Off-nadir angle 0°		Off-nadir angle 15°		Off-nadir angle 30°	
	Accuracy	Computation time [ms]	Accuracy	Computation time [ms]	Accuracy	Computation time [ms]
PECAN	0.9821	9800.1	0.9961	9973.9	0.9995	12804.2
CITRA	0.4362	52.0	0.5073	61.0	0.4178	89.5
CIGNN	0.6652	2.191	0.6971	2.076	0.7048	2.231
CIGNN + PECAN	0.9018	20.138	0.9455	25.685	0.9622	49.844

* Best: Green, Second-best: Light Green

Table 1. **Per-scenario comparison of accuracy and computation time**. PECAN reports the highest accuracy, while CIGNN has the shortest computation time. CIGNN+PECAN seems to be a compromise between CIGNN and PECAN as its performance is the second-best across all scenarios.

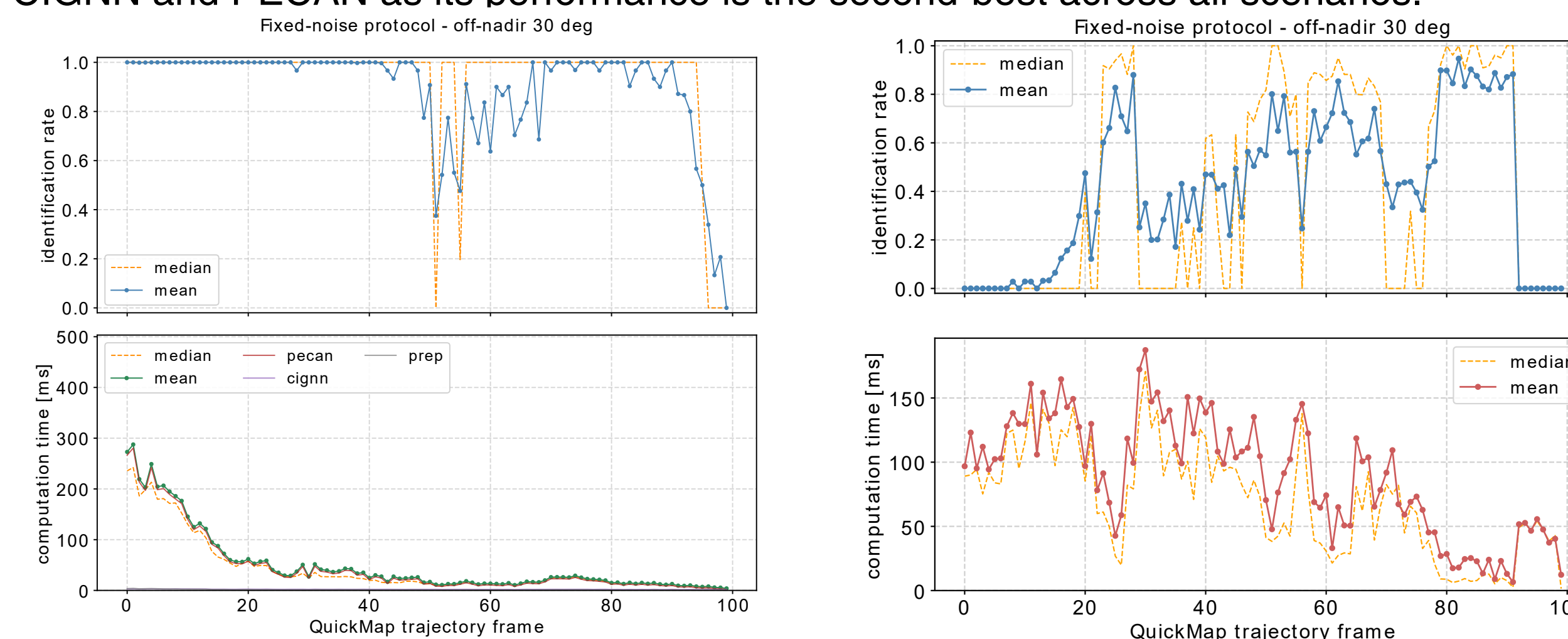


Fig. 4. **Per-frame accuracy and computation time of CIGNN(left) and CITRA(right)**. CIGNN performs worse when images contain fewer craters, while CITRA performs better on such images.

Conclusion

• Limitations

1. **Crater detection is bypassed**. We instead apply false-positive, false-negative, size, and position noise following Rodda et al.
2. **Small Comparison Group**. We compared CIGNN and CITRA with PECAN only. Other state-of-the-art methods such as CraterSense should be compared.

• Conclusion

- **SID-to-CID transfer is effective**. CIGNN and CITRA successfully adapt modern SID concepts to CID.
- **Real-Time Potential**. The results indicate strong potential for real-time lunar lander localization with its computation times below 0.1s.

• Future Work

1. **Integration of Crater Detection**. We plan to integrate crater detection methods using Ellipse R-CNN and VLM-based approaches.
2. **Robust Comparison with Existing Methods**. We plan to add more open-sourced CID methods and evaluation metrics (e.g., F1 score and pose error).